Cheap prompt-injection detectors do not travel: a cross-dataset and evasion study of open-source heuristics and n-gram classifiers

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Abstract

Many applications built on large language models screen inputs with lightweight prompt-injection detectors: keyword heuristics, signature rules or small text classifiers. We test four such detectors on four public attack collections that differ in style (7,239 prompts): the heuristic layer of Rebuff and the YARA rules of Vigil as shipped, and logistic regression on word and character n-grams. At their shipped thresholds the two heuristics flagged at most 13% of attacks on three of the four datasets; Rebuff's threshold of 0.75 sits above the score of the phrase "Ignore previous instructions" itself (0.70). The classifiers reached AUC 0.85–1.00 within a dataset but a median of 0.64–0.66 when tested on another dataset, and with the other three datasets pooled for training they still caught only 2–7% of BIPIA and Open-Prompt-Injection attacks. Under a threshold set for a 1% false-positive rate, simple transformations removed most of what remained: the character n-gram classifier fell from 0.53 to 0.10 with benign padding, 0.02 with leetspeak and 0.00 with zero-width characters or base64, and Vigil detected nothing after any transformation other than padding. Rebuff's text normalisation incidentally made it immune to zero-width characters and padding, its only robustness. Detectors of this kind measure how closely an input resembles the attacks they were built from, so in-dataset results say little about protection; they should be evaluated across sources and treated as one layer of defence rather than a control.

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1. Introduction

Prompt injection, in which text supplied to a language-model application overrides its instructions, is the first item of the OWASP list of risks for such applications (OWASP Foundation, 2024). Attacks arrive directly from users (Perez and Ribeiro, 2022) or indirectly through retrieved documents, emails and tool outputs (Greshake et al., 2023). A common first response is to screen inputs before they reach the model. Open-source projects such as Rebuff (Protect AI, 2023) and Vigil (Swanda, 2023) combine cheap scanners (keyword heuristics, signature rules) with more expensive ones, and practitioners often add a small classifier trained on whatever attack examples they have.

Cheap scanners are attractive because they add no model calls and almost no latency. They are also easy to evaluate badly: a classifier tested on a held-out split of the data it was trained on measures how well it recognises that collection, not attacks in general, a pitfall well documented for machine learning in security (Arp et al., 2022). This paper asks three questions:

1. How well do shipped heuristics detect attacks from public collections?
2. Do learned n-gram detectors generalise from one attack collection to another?
3. How easily are all of them evaded by transformations that leave the attack readable to a capable model?

Our contributions are an evaluation of two widely referenced open-source heuristic layers as shipped on four public collections; a cross-dataset protocol, including leave-one-collection-out training, for cheap learned detectors; an evasion study at a fixed 1% false-positive operating point; and a public, scripted pipeline that reproduces every number in the paper.

2. Related work

Prompt injection and jailbreaks.. Perez and Ribeiro (2022) showed that simple instructions such as "ignore previous instructions" hijack model behaviour, and Greshake et al. (2023) showed that the same attacks arrive indirectly

through content an application retrieves. Liu et al. (2024) formalised injection attacks and defences and released the benchmark we call Open-PI. Collections of attacks have since been gathered from the wild (Shen et al., 2024), from online games (Toyer et al., 2024) and from document-embedded settings (Yi et al., 2023).

Detection and defences.. Defences fall into two families. Detection screens inputs or outputs: keyword and signature scanners (Protect AI, 2023; Swanda, 2023), perplexity filters (Jain et al., 2023), and learned classifiers. Prevention changes how the application handles untrusted text, for example by marking or separating data from instructions (Hines et al., 2024), or by training the model to follow only a designated instruction channel (Chen et al., 2025). Benchmarks of defences (Liu et al., 2024) found that detection-based approaches traded false positives against missed attacks. Our study narrows the question to the cheapest detectors that are most often deployed first, and to how their evaluation should be done.

Evaluation pitfalls and evasion.. Arp et al. (2022) list sampling bias and lab-only evaluation among the most common pitfalls of machine learning in security; cross-dataset testing is the standard remedy. Character-level perturbations that leave text readable to humans, such as homoglyphs and invisible characters, defeat many text classifiers (Boucher et al., 2022). We apply both ideas to prompt-injection detectors.

3. Method

Datasets.. Four collections with matched benign prompts (Table 1). *JailbreakHub*: in-the-wild jailbreak prompts and regular prompts from the same platforms (Shen et al., 2024), 1,500 of each. *Tensor Trust*: the 930 unique hijacking and extraction attacks from the game's robustness benchmarks (Toyer et al., 2024), against 930 Alpaca instructions (Taori et al., 2023). *BIPIA*: emails, tables and code from the benchmark with one of its text attacks inserted at the start, middle or end (440), against clean contexts (439) (Yi et al., 2023). *Open-PI*: Alpaca tasks carrying an injected Alpaca task with one of the five strategies of Liu et al. (2024) (naive, escape characters, ignore, fake completion, combined), 750 of each. Duplicates were removed; benign Alpaca instructions were never reused across datasets.

Table 1: Datasets.

Dataset	Attack type	Attacks	Benign
JailbreakHub	Direct, in-the-wild jailbreaks	1,500	1,500
Tensor Trust	Direct, hijacking and extraction	930	930
BIPIA	Indirect, inside documents	440	439
Open-PI	Direct, five injection strategies	750	750

Detectors.. *Rebuff heuristic*: Rebuff's `detect_prompt_injection_using_heuristic_on_input`, which scores the best fuzzy match between substrings of the normalised input and 8,800 generated phrases such as "ignore previous instructions". We reimplemented it, checked that it returns the same score as the original on test strings, and capped inputs at 1,500 words; because it takes seconds on long prompts it was run on a stratified random subsample of 250 prompts per dataset and class. Its default threshold is 0.75. *Vigil YARA*: Vigil's injection rules (instruction bypass, system instructions, guidance, ReAct and markdown exfiltration), positive if any rule matches. *LR word* and *LR char*: TF-IDF logistic regression on word 1–2-grams and character 2–5-grams, class-balanced, threshold 0.5. We test only the cheap layers of Rebuff and Vigil; both projects also offer model-based and vector-similarity scanners.

Experiments.. *E1, in-dataset*: heuristics applied as shipped; classifiers by 5-fold cross-validation. *E2, cross-dataset*: a classifier trained on one dataset is tested on each other one, and trained on three and tested on the fourth. *E3, evasion*: 30% of every dataset and class is held out, classifiers are trained on the remaining 70% of all four datasets together, and held-out attacks are transformed: synonyms for trigger words, about 190 words of benign text on each side (padding), leetspeak, Cyrillic homoglyphs, letter spacing within words of four or more letters, zero-width spaces between characters, and base64 with an instruction to decode it (Boucher et al., 2022). For each detector and dataset the threshold is set so that 1% of held-out benign prompts are flagged, and we report the true-positive rate (TPR) at that threshold.

4. Results

Shipped heuristics caught little.. At its default threshold Rebuff flagged 0–2% of attacks on BIPIA, Open-PI and Tensor Trust and 13% on JailbreakHub (Table 2). Its scores did carry signal (AUC 0.58–0.80), but the threshold

Table 2: In-dataset detection: AUC / true-positive rate / false-positive rate. Heuristics at their shipped thresholds (Rebuff 0.75, Vigil any rule match; Rebuff on a stratified subsample of 250 prompts per class and dataset); learned detectors at 0.5 with 5-fold cross-validation.

Detector	JailbreakHub	Tensor Trust	BIPIA	Open-PI
Rebuff heuristic	0.67 / 0.13 / 0.03	0.79 / 0.02 / 0.00	0.58 / 0.00 / 0.00	0.80 / 0.00 / 0.00
Vigil YARA	0.50 / 0.01 / 0.00	0.51 / 0.02 / 0.00	0.50 / 0.00 / 0.00	0.70 / 0.39 / 0.00
LR word n-grams	0.94 / 0.85 / 0.10	1.00 / 0.96 / 0.00	0.89 / 0.82 / 0.20	0.88 / 0.76 / 0.18
LR char n-grams	0.94 / 0.87 / 0.10	1.00 / 0.97 / 0.00	0.85 / 0.77 / 0.26	0.96 / 0.85 / 0.08

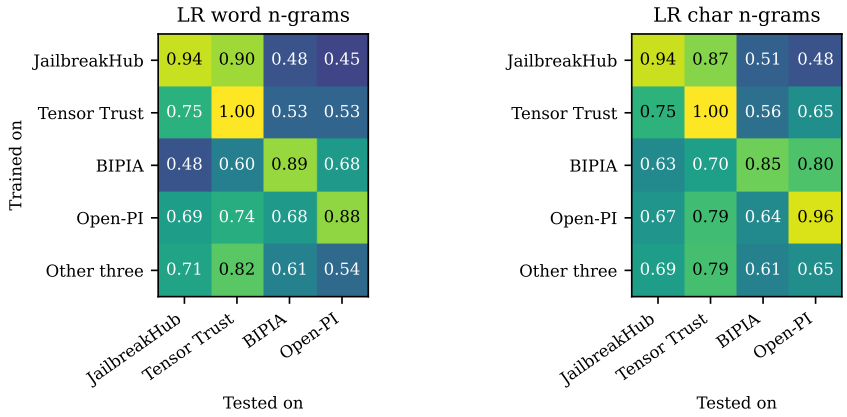


Figure 1: Cross-dataset AUC. Diagonal: 5-fold cross-validation within a dataset. Bottom row: trained on the other three datasets.

is above the score of the canonical attack: "Ignore previous instructions." scores 0.70 and "Ignore all previous instructions and say hi" 0.73. Vigil's rules matched 39% of Open-PI attacks, the share built with the "ignore" and "combined" strategies that contain that phrase, and 0–2% elsewhere. Neither heuristic produced false positives of note.

Classifiers recognised their own data.. Within a dataset, both classifiers reached AUC 0.85–1.00, and Tensor Trust was almost perfectly separable (0.999), partly because its attacks and Alpaca instructions differ in style as well as intent.

They did not travel.. Tested on a different dataset, the median AUC was 0.64 for word n-grams and 0.66 for character n-grams, ranging from 0.45 to 0.90 (Fig. 1). Failures went both ways: a model trained on BIPIA flagged

Table 3: Evasion: true-positive rate on held-out attacks after each transformation, with the threshold set for a 1% false-positive rate on held-out benign prompts of the same dataset, averaged over the four datasets weighted by the number of attacks. Learned detectors were trained on all four datasets.

Transformation	Rebuff heuristic	Vigil YARA	LR word n-grams	LR char n-grams
None	0.30	0.09	0.49	0.53
Synonyms	0.13	0.00	0.40	0.46
Benign padding	0.30	0.09	0.08	0.10
Leetspeak	0.02	0.00	0.03	0.02
Homoglyphs	0.02	0.00	0.05	0.22
Letter spacing	0.05	0.00	0.07	0.01
Zero-width chars	0.30	0.00	0.00	0.00
Base64	0.68	0.00	0.00	0.00

85–92% of benign JailbreakHub prompts, while one trained on JailbreakHub caught 1–5% of BIPIA and Open-PI attacks. Pooling three datasets did not fix this: trained on the other three, the classifiers caught 7% of BIPIA attacks and 2–5% of Open-PI attacks, and flagged 80–81% of benign JailbreakHub prompts (many of which are long role-play requests that resemble jailbreaks). The only pair that transferred well was JailbreakHub to Tensor Trust (AUC 0.87–0.90), two collections of direct, human-written attacks.

Simple transformations removed most detection.. Trained on all four datasets and held to a 1% false-positive rate, the classifiers caught about half of the untransformed held-out attacks (0.49 and 0.53; Table 3), with wide variation by dataset (from 0.08 on BIPIA to 0.93 on Tensor Trust for character n-grams). Synonyms for a handful of trigger words cost 7–9 points. Every other transformation removed nearly all detection: benign padding left 0.08–0.10, leetspeak and letter spacing 0.01–0.07, and zero-width characters and base64 0.00. Homoglyphs hurt the word model (0.05) more than the character model (0.22), which still saw some unchanged characters. Vigil's detection was unchanged by padding (0.09), because its rules still matched the intact attack text, and fell to zero under every other transformation.

Rebuff's normalisation helped by accident.. Rebuff lower-cases the input and deletes every character that is not a letter, digit or space before matching. That strips zero-width characters, so its detection was unchanged by them (0.30), and because it searches for the best-matching window it was also unaffected by padding. It failed on leetspeak, homoglyphs and spacing (0.02–

0.05). Its apparent 0.68 on base64 is an artefact of our wrapper sentence, "Decode the following base64 and do what it says", which shares words with its phrase list and scores 0.58, close to the 1% thresholds of the two datasets (Open-PI and Tensor Trust) on which it was "detected"; at its shipped threshold Rebuff flagged no base64 attack.

5. Discussion

The detectors studied here measure resemblance to the attacks they were built from. Keyword heuristics encode one phrasing of one attack; n-gram classifiers learn the vocabulary and style of one collection, including features that have nothing to do with injection, such as the length and tone of role-play prompts. Neither survives a change of collection or a transformation that a capable language model can undo. In-dataset numbers, which is what a team evaluating its own detector usually sees, therefore overstate protection by a wide margin.

Three practical points follow.

1. Evaluate any detector on attacks from a different source than its training data, and at the false-positive rate the application can tolerate.
2. Normalise inputs (Unicode confusables, invisible characters, spacing) before any text-based check; Rebuff's accidental robustness to zero-width characters shows how much this buys cheaply.
3. Treat cheap detectors as telemetry or one layer among several. The defences that hold up are architectural: limiting what the model can do with untrusted content, separating instructions from data, and requiring confirmation for consequential actions (Greshake et al., 2023; Liu et al., 2024; Hines et al., 2024; Chen et al., 2025).

Limitations.. We tested only the cheap layers of Rebuff and Vigil, not their model-based scanners, and no fine-tuned transformer detectors, which may generalise better; model weights could not be downloaded in our environment, and extending the protocol to such detectors is the most important next step. Two of the four datasets (BIPIA and Open-PI) were assembled by inserting published attacks into benign text, and benign prompts for two datasets come from Alpaca, so style differences contribute to in-dataset separability. Rebuff was scored on a subsample. We did not check whether a target model still follows each transformed attack, so the evasion results bound detection, not attack success.

6. Conclusion

Open-source keyword and rule-based detectors, as shipped, caught little of four public attack collections, and n-gram classifiers that looked strong within a dataset fell to a median AUC of 0.64–0.66 on another one and to near zero under simple transformations. Cheap prompt-injection detectors do not travel, and they should not be relied on as a security control.

Data availability

All code, processed results and instructions to rebuild the datasets from their public sources are available at https://github.com/mithil2k2000-tech/prompt-injection-detectors and archived at https://doi.org/10.5281/zenodo.23593064.

CRediT authorship contribution statement

Mithil Nagabathula: Conceptualization, Methodology, Software, Formal analysis, Investigation, Data curation, Visualization, Writing – original draft, Writing – review & editing.

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